

Loudness budget constraints on high-frequency amplification in precipitous hearing loss

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Abstract

Background: In precipitous high-frequency hearing loss, achieving sufficient audibility in the high frequencies is notoriously difficult. Historically, this limitation has been attributed to the choice of prescriptive fitting rationale or to inherent physiological damage.

Purpose: This study examines an additional, structural explanation based on loudness budget arithmetic: unamplified speech energy in the residual normal-hearing low frequencies consumes much of the acceptable loudness budget before any gain is prescribed, reducing the capacity available to restore high-frequency audibility.

Research Design: A computational modeling study using the AUDMOD specific-loudness model alongside the Speech Intelligibility Index (SII).

Study Sample: N/A (computational modeling of seven canonical audiometric profiles and sixteen synthetic audiograms).

Data Collection and Analysis: The loudness budget was decomposed for seven canonical profiles at 50, 65, and 80 dB SPL. A systematic sweep then crossed four audible edge frequencies with four high-frequency slopes, evaluated at four loudness budgets, three desensitization scales, and two minimum-gain floors (0 and -10 dB), optimizing each floor separately and comparing solutions at matched loudness. Finally, an iso-loudness control constrained a speech intelligibility maximizer to NAL-NL2's own achieved loudness for each profile, isolating the cost of the 0 dB insertion floor.

Results: For profile A4 at 65 dB SPL, unamplified speech consumes 64% of the normative loudness ceiling before any prescriptive gain is applied. Relaxing the minimum-gain floor from 0 to -10 dB improved the desensitized SII by 0.038 to 0.079 where the audible edge lay at 1000 Hz and by 0.009 to 0.023 at 1500 Hz, against an optimizer resolution limit of 0.007; at 2000 and 3000 Hz the benefit was not distinguishable from solver noise. At matched loudness, the freed capacity was drawn from the low frequencies and from 8 kHz and spent in the 2–4 kHz region. In the iso-loudness control the floor effect was largest for the moderate sloping profile A3 (+0.073) and smaller for the precipitous profiles A4 (+0.020) and A5 (+0.009), shrinking further as the desensitization penalty strengthened.

Conclusions: Loudness crowding-out is a general structural boundary for profiles combining robust low-frequency hearing with impaired high frequencies, but its measurable benefit is confined to losses with a relatively low audible edge and is progressively discounted by physiological desensitization. Realizing it requires negative low-frequency insertion gain (attenuation below the unaided ear-canal level), achievable only with an occluding earmold that these listeners are unlikely to tolerate — which is an argument for frequency-lowering signal processing rather than for more high-frequency gain.

Clinical Relevance Statement: Clinicians who cannot fully restore high-frequency audibility in sloping or precipitous losses should recognize that a structural loudness ceiling, not necessarily an inadequate fitting, may be limiting the achievable gain, particularly when low-frequency hearing is largely spared. Because realizing more of this budget requires occluding the ear canal, which produces an occlusion effect these patients are unlikely to tolerate, approaches that deliver high-frequency cues where hearing is better, such as frequency lowering, may warrant consideration over further high-frequency gain.

Introduction

In the fitting of precipitous or profound high-frequency hearing loss, achieving adequate high-frequency audibility without exceeding normative loudness targets is a central clinical challenge. Historically, the failure to restore high-frequency audibility has been attributed to the inherent physiological damage of the auditory periphery—specifically,

hearing-loss desensitization and dead regions (Moore 2001), which render severe high-frequency amplification effectively useless for speech recognition (Ching, Dillon, and Byrne 1998; Hogan and Turner 1998). Consequently, modern prescriptive rationales like NAL-NL2 (Keidser et al. 2011) deliberately limit high-frequency gain for profound thresholds to avoid prescribing “wasted” amplification.

While desensitization dictates the *utility* of high-frequency gain, it is proposed that the fundamental arithmetic of broadband loudness summation acts as a parallel, independent structural constraint on the *capacity* to provide that gain. For a given audiogram and input level, unamplified speech inherently produces a baseline quantity of loudness (L_0) through residual hearing. If a patient’s overall target loudness is constrained by a normative ceiling (L_{cap}), the model-predicted loudness budget available to “purchase” high-frequency audibility via prescriptive gain is defined as $L_{cap} - L_0$. Because this relationship is governed entirely by the auditory periphery and the acoustic speech spectrum, it acts as a fundamental boundary condition that profoundly constrains loudness-based rationales (e.g., NAL-NL2) and any theoretical attempt at intelligibility maximization.

While patient preference and some audiological paradigms often favor preserving or amplifying low-frequency cues in profound loss because they are the only remaining regions of robust functional hearing, the standard ANSI S3.5-1997 Speech Intelligibility Index (SII) (ANSI 1997) assigns the bulk of its importance weighting to the 1-4 kHz speech bands, with relatively little weight applied below 500 Hz. Consequently, an unconstrained intelligibility-maximizing algorithm operating under a strict loudness budget will systematically attenuate low-frequency audibility, as well as audibility at 8 kHz where the importance weighting is likewise low, to fund high-frequency gain. While it is a mathematical axiom that relaxing a constraint in an optimization algorithm yields a higher theoretical optimum, the clinical relevance lies in quantifying the sheer magnitude of this trade-off for realistic audiometric profiles. By framing this behavior as an efficiency statement relative to an absolute budget, this analysis quantifies the exact degree to which “broadband loudness crowding-out” limits high-frequency restoration, investigating whether it is uniquely tied to precipitous losses or represents a general property of near-normal low-frequency hearing.

Methods

The AUDMOD Specific-Loudness Model

To isolate the model-predicted loudness constraints operating on the speech spectrum, the canonical AUDMOD specific-loudness model was utilized. This model converts acoustic excitation patterns into specific loudness (sones/ERB) and integrates them across the frequency spectrum to predict overall broadband loudness. By benchmarking the raw, unamplified long-term average speech spectrum (LTASS) (Byrne et al. 1994) through this model using the seven standard hypothetical audiometric profiles established by Johnson and Dillon (Johnson and Dillon 2011), the baseline loudness (L_0) natively generated by residual hearing was established.

Study Sample and Audiometric Profiles

The study utilizes seven canonical audiometric profiles (A1–A7) representing standard clinical configurations (Table 1). For profiles A6 (Mixed) and A7 (Conductive), the conductive air-bone gap component is modeled in the AUDMOD specific-loudness engine as a linear attenuation block that strictly subtracts from the aided speech spectrum before it enters the cochlear simulation.

Table 1. Canonical Audiometric Profiles (dB HL)

Pro- file	Description	250	500	1000	2000	4000	8000	ABG
A1	Mild	15	20	30	40	50	60	0
A2	Reverse slope	60	50	40	30	20	15	0
A3	Moderate sloping	10	20	40	50	55	60	0
A4	Mod-severe precip.	0	0	10	40	70	80	0
A5	Profound precip.	10	10	20	60	80	100	0
A6	Mixed	50	55	60	65	75	80	30 dB
A7	Conductive	50	50	50	50	50	50	50 dB

The Open-NL Computational Instrument

To compute the theoretical limits of high-frequency amplification and map the boundary of achievable gain, Open-NL, a modular computational testbed, was employed. Open-NL couples a multi-start Nelder-Mead SII optimizer to the specific-loudness engine. Rather

than serving as a clinical prescription, Open-NL functions here strictly as an analytical instrument.

For the purposes of this study, all input speech was calibrated to the standard ANSI S3.5-1997 Long-Term Average Speech Spectrum (LTASS) (ANSI 1997). Because the analysis focuses entirely on average conversational speech, the Wide Dynamic Range Compression (WDRC) capabilities of Open-NL were bypassed, reducing the optimizer to a single-level linear insertion gain solver at 65 dB SPL.

The optimizer was subjected to a global minimum insertion gain constraint across all frequency bands (the `vent_floor` parameter). A negative insertion gain floor represents the passive insertion loss of an occluding earmold (the Real-Ear Occluded Response minus the Real-Ear Unaided Response). Measured insertion loss is inherently frequency-dependent and is typically smallest at the lowest frequencies, even for completely unvented molds (Kuk, Keenan, and Lau 2009). The flat -10 dB floor used here is therefore an idealization that is likely optimistic in the lowest frequency bands, rather than a model of any specific earmold. While mathematically enforced as a global floor, for sensorineural profiles with sloping configurations the floor binds only at the low frequencies, with A2, a reverse-slope profile, the exception. Baseline clinical targets were generated using NAL-NL2 Version 2 software (Keidser et al. 2011) at 65 dB SPL for a symmetrical, bilateral fitting for an experienced user of unknown gender. To isolate the rationale’s fundamental prescription for the audiogram from the secondary acoustic effects of venting, the targets were intentionally generated using a fully occluded (#13 tubing) coupling rather than an open fitting. While an open fitting is the clinical standard of care for these profiles, generating NAL-NL2 targets with an open coupling instructs the software to automatically cut low-frequency prescribed gain. Utilizing an occluded setting ensures the baseline targets reflect the rationale’s true theoretical intent. However, it must be noted that because NAL-NL2 was parameterized for a bilateral fitting, the rationale automatically applied a binaural loudness correction that systematically lowers prescribed gain. Because the computational instruments utilized in this analysis are strictly monaural, this introduces a methodological mismatch. While this binaural correction contributes slightly to NAL-NL2’s lower overall predicted loudness, the magnitude of the discrepancy is small relative to the structural boundaries of the $L_{cap} - L_0$ budget.

C++ Instrument Validation

Because Open-NL evaluates tens of thousands of candidate gain curves during optimization, evaluating loudness via external Python or MATLAB calls introduces prohibitive computational overhead. To achieve microsecond evaluation times, the AUDMOD specific-loudness engine (AMT 1.6.0 (Majdak, Hollmach, and Baumgartner 2022), bramslow2004) was ported entirely to C++ using Rcpp.

The C++ engine was validated at three distinct levels: port fidelity, input path consistency, and absolute reference-free behavior.

First, port fidelity was established by comparing the C++ engine to the AMT bramslow2004 (AUDMOD) implementation stage-by-stage. Across 23 test cases, the maximum relative discrepancy at any stage of computation was 5.71×10^{-15} . This comparison establishes that the C++ port reproduces AUDMOD; it does not bear on whether AUDMOD itself is accurate.

Second, the input path for spectrum construction feeding the engine was verified. The two independent routines used to compute normal-hearing speech loudness in this work return identical values at 50, 65, and 80 dB SPL (yielding 3.1577, 9.0333, and 22.3887 sones, respectively). Evaluating the 65 dB SPL speech spectrum on a 10–23990 Hz grid without renormalization yields 9.0333 sones, whereas evaluation on a 20–15000 Hz grid with renormalization yields 9.0317 sones; this difference of 0.0016 sones demonstrates that grid range and renormalization are immaterial. Furthermore, supplying a precomputed reference versus letting the wrapper build one dynamically yields identical results for profiles A1 (13.2297 sones), A4 (20.7120 sones), and A5 (16.6072 sones).

Third, the absolute behavior of the model was verified using reference-free checks, as the port and AMT would share any error. Loudness correctly evaluates to zero below threshold: for normal hearing, a 1 kHz tone remains silent up to +5 dB SPL, and for profile A5, 4 kHz remains silent up to 60 dB SPL against an 80 dB HL threshold. Recruitment is present and ordered by threshold at 4 kHz across profiles A1–A5; the apparent non-monotonicity across profiles reflects the audiometric configuration, not severity (e.g., A2 is a reverse-slope profile with a 20 dB HL threshold at 4 kHz). In normal hearing, the loudness growth for a 1 kHz tone is 1.75 to 1.91 per 10 dB above 40 phons, which

characterizes the model as mildly compressive relative to the textbook expectation of a doubling per 10 dB.

A notable limitation is that the absolute sone scale is inherited from AUDMOD and has not been independently verified against loudness judgments. An attempt to check the 1 kHz / 40 dB SPL = 1 sone anchor was inconclusive: the R wrapper interpolates the input spectrum linearly in dB across a log-frequency grid, which is appropriate for the smooth, densely sampled speech spectra used throughout this work but cannot faithfully represent a pure tone. The smooth-spectrum control was stable at 9.0333 sones across every input grid spacing tested, ensuring the analyses reported here are unaffected. We do not claim the absolute scale is either correct or incorrect.

Optimization Procedure

The underlying optimization routine utilized a multi-start Nelder-Mead simplex algorithm over six free parameters: a gain shift at each audiometric octave frequency from 250 to 8000 Hz, applied on top of the Open-NL heuristic prescription and interpolated logarithmically across the 21 analysis bands. All Open-NL runs used the package defaults for the heuristic prescription: a fully occluded coupling with no vent correction, the severe-loss booster disabled, and no demographic adjustments.

The solutions generated are penalized-optimal rather than strictly SII-optimal. Specifically, the optimizer maximizes a nine-term penalized objective function, not raw SII. The objective consists of the SII (scaled by 100) minus eight penalty terms: an anchor penalty ($\text{sum}(\text{abs}(\text{shifts})) * 0.1$) which acts as an L1 shrinkage toward the Open-NL heuristic prescription, a roughness penalty ($\text{sum}(\text{diff}(\text{gain_oct})^2) * 0.001$), a loudness hinge penalty scaled at 2000 per sone over the cap, and additional out-of-bounds, SPL, order, compression ratio (CR), and air-bone gap (ABG) terms (see R/open_nl.R lines 268-284).

To mitigate local minima, multi-start initialization was employed across a bounded parameter grid, and the solver retains the highest value of the penalized objective, not the highest SII. Because the objective includes the anchor penalty (an L1 shrinkage toward the heuristic prescription) and the roughness penalty, additional restarts improve the objective and can return solutions with lower SII.

Additionally, the number of random restarts (`open_nl_starts`) is a fixed setting (set to 20) whose value alters the prescribed gains due to the non-convex parameter space. (Note that because the random number generator seed is deterministic based on the audiogram and level, repeated runs are bit-identical and cannot be used to estimate variability.)

Results

Loudness Budget Decomposition

To establish the model-predicted loudness constraints operating on high-frequency amplification, the unamplified loudness (L_0) produced by speech across three conversational input levels (50, 65, and 80 dB SPL) was first computed. This baseline was then compared against a normative broadband loudness ceiling (L_{cap}) for each level.

To avoid arbitrary definitions of tolerance, the normative target ceilings (L_{cap}) are strictly defined as the normal-hearing loudness of unaided speech at the evaluation level. When the 50, 65, and 80 dB SPL long-term average speech spectra are processed through the specific-loudness engine for a 0 dB HL profile, the model predicts overall loudnesses of 3.16, 9.03, and 22.39 sones, respectively. These ceilings are identical for all listeners and are not listener-specific. The difference ($L_{cap} - L_0$) therefore represents the remaining model-predicted loudness budget available to “purchase” high-frequency audibility via prescriptive gain before exceeding normal-hearing loudness limits.

An empirical anchor for this ceiling would be preferable to a normative one, so we tested whether an established rationale provides one. It does not: NAL-NL2-prescribed aided loudness for 65 dB SPL speech, expressed as a fraction of the normative ceiling, scatters by nearly a factor of two with no consistent value (0.46 for A1, 0.39 for A2, 0.40 for A3, 0.73 for A4, and 0.61 for A5). Because NAL-NL2 optimizes loudness against its own criteria rather than targeting a fixed proportion of normal-hearing loudness, it cannot be used to calibrate this parameter.

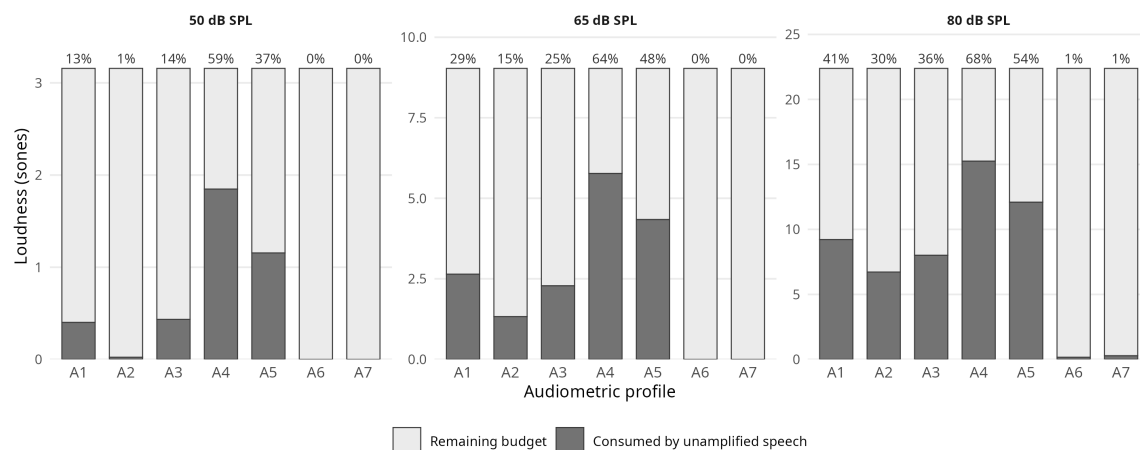
L_{cap} is therefore a modelling assumption, not a measured quantity. Because true loudness discomfort or target loudness can vary significantly across individual patients and fitting rationales, analyzing a single fixed normative cap for 65 dB SPL speech (e.g., 9.03 sones) fails to capture the full optimization boundary. This is why the audiogram

family sweep evaluates four different loudness budgets per listener rather than a single fixed value: the sweep across budgets is the sensitivity analysis for this assumption, and conclusions that hold across the range do not depend on the particular value chosen.

Table 2 presents this decomposition for the seven canonical audiometric profiles. For mild (A1) or moderate sloping (A3) profiles, unamplified speech consumes a minority of the normative loudness budget across all levels. This leaves ample capacity for prescriptive algorithms to apply positive insertion gain across the frequency spectrum.

Table 2. Loudness Budget Decomposition (Sones) across Audiometric Profiles at 50, 65, and 80 dB SPL

Profile	Description	L_0 (50)	L_{cap} (50)	L_0 (65)	L_{cap} (65)	L_0 (80)	L_{cap} (80)
A1	Mild	0.40	3.16	2.65	9.03	9.22	22.39
A2	Reverse Slope	0.02	3.16	1.33	9.03	6.72	22.39
A3	Moderate Sloping	0.43	3.16	2.29	9.03	8.01	22.39
A4	Mod-Severe Precipitous	1.85	3.16	5.77	9.03	15.25	22.39
A5	Profound Precipitous	1.16	3.16	4.34	9.03	12.10	22.39
A6	Mixed	0.00	3.16	0.00	9.03	0.17	22.39
A7	Conductive	0.00	3.16	0.00	9.03	0.28	22.39



Profiles as defined in Table 1. Each bar is the normative ceiling for that input level, identical across listeners. The dark segment is the loudness of unamplified speech generated by residual hearing; the light segment is what remains to purchase high-frequency audibility. Percentages give the fraction of the ceiling consumed before any gain is prescribed.

Figure 1: Fraction of the normative loudness ceiling consumed by unamplified speech, for the seven canonical profiles (defined in Table 1) at input levels of 50, 65, and 80 dB SPL. The ceiling is the loudness of unaided speech for a listener with typical hearing at that level, and is identical across profiles. The remainder of each bar’s height to 100% is the budget available to purchase high-frequency audibility before any gain is prescribed. However, a structural bottleneck emerges in precipitous profiles (A4 and A5) across the entire dynamic range. As illustrated in Figure 1, the fraction of the loudness budget consumed by unamplified speech rises with input level for every profile, so the constraint tightens as speech gets louder. Profile A4 consumes 58.5%, 63.9%, and 68.1% of the ceiling at 50, 65, and 80 dB SPL respectively—roughly twice the fraction consumed by the mild (A1: 12.7%, 29.3%, 41.2%) and moderate sloping (A3: 13.7%, 25.3%, 35.8%) profiles, and substantially more than the reverse slope profile (A2: 0.7%, 14.7%, 30.0%). Profile A5 consumes a smaller fraction than A4 (36.6%, 48.1%, 54.0%) despite being the more profound loss; this is because A5’s high-frequency thresholds are so elevated that unamplified speech is largely inaudible there and contributes little loudness. Crucially, if a prescriptive formula enforces a rigid low-frequency insertion gain floor of 0 dB (preventing attenuation), these L_0 values cannot be reduced. What remains is thus far short of the immense high-frequency gain required to cross the profound high-frequency thresholds, and high-frequency audibility is systematically crowded out by residual low-frequency hearing at all input levels.

The Audiogram Family Sweep

A defect in the `open_nl()` routine caused an earlier version of this minimum-gain floor contrast to report a rectified difference rather than a true effect. When called with a negative insertion floor, the function executed a dual-run branch that evaluated the optimization at both the requested floor and a strict 0 dB floor, automatically returning whichever yielded the higher raw speech intelligibility index—meaning branch selection used raw SII, whereas selection within each branch used the penalized objective (see `R/open_nl.R` lines 381–395 for 65 dB SPL and 404–418 for other levels). Consequently, the published difference between the –10 dB and 0 dB floors was not $SII(-10) - SII(0)$ but $\max(0, SII(-10) - SII(0))$. Across all 192 paired cells of the original audiogram family analysis, not one difference took a negative value; this is impossible in a genuine comparison because the 0 dB feasible set is contained within the –10 dB feasible set, meaning negative differences must occur wherever the true effect is small relative to optimizer precision. This bounding explains why 63 of the 192 paired cells appeared as exactly zero. These zeros were simply 0 dB solutions differenced against themselves rather than legitimate null results. Their frequency rose with the audible edge frequency, from 31% of cells at 1000 Hz to 61% at 3000 Hz, consistent with the 0 dB branch winning most often where relaxing the floor helps least. Because this defect was isolated to branch selection and did not compromise the underlying loudness engine or the optimizer itself, the analysis code for this paper disables the dual-run branch so that each floor is optimized exactly once at the floor requested (the original package behavior is documented in a separate issue). The affected results have been recomputed under this corrected methodology and are reported below as side-by-side solutions with their achieved loudness.

Sixteen synthetic audiograms crossing four audible edge frequencies (1000, 1500, 2000, 3000 Hz) with four high-frequency slopes (20, 30, 40, 50 dB/octave) were evaluated at four loudness budgets (0.5, 1, 2, and 3 sones above each listener’s own unaided loudness), three desensitization scales (0, 0.5, 1), and two minimum-gain floors (0 and –10 dB). Each floor was optimized once at the floor requested with 20 random restarts, and solutions at the two floors are reported side by side with their achieved loudness rather than as a difference.

In 133 of the 192 paired cells, the two solutions differ in achieved loudness by no more than 0.001 sones, so the comparison is like-for-like at matched loudness; the cells that fail this test are concentrated at the loosest budget, where the loudness cap stops binding and the optimizer no longer spends the capacity available to it.

A negative difference between floors is impossible in a genuine comparison, since the 0 dB feasible set is contained within the -10 dB set. Across the 133 paired cells at matched loudness, 46 such excursions occur; their median magnitude is 0.0009 and the largest is 0.0074. We take the maximum as the optimizer's resolution limit, so differences below roughly 0.007 are not distinguishable from solver noise.

At the tightest budget, the mean SII change from relaxing the floor is 0.079, 0.054, and 0.038 at a 1000 Hz audible edge for desensitization scales 0, 0.5, and 1 respectively; 0.023, 0.016, and 0.009 at 1500 Hz; 0.010, 0.008, and 0.005 at 2000 Hz; and approximately zero at 3000 Hz. The effect declines monotonically with desensitization scale and with loudness budget. It is unambiguous at 1000 and 1500 Hz, comparable to the resolution limit at 2000 Hz, and indistinguishable from zero at 3000 Hz. At the tightest budget and a 1000 Hz audible edge, averaged across desensitization scales, prescribed gain falls by 6.8 dB at 500 Hz, 3.3 dB at 1000 Hz, and 7.9 dB at 8 kHz, and rises by 10.0 dB at 2000 Hz and 6.4 dB at 4000 Hz. Because the paired solutions are at matched loudness, these changes trade off against one another: capacity is drawn from the near-normal low frequencies and from 8 kHz, and spent in the 2-4 kHz region. The reduction at 8 kHz reflects two features of the implementation. The SII assigns little importance to that region, and the loudness model extrapolates the aided spectrum above 8 kHz at -24 dB per octave from the 8 kHz value, so gain there incurs loudness from a spectral region the SII does not score.

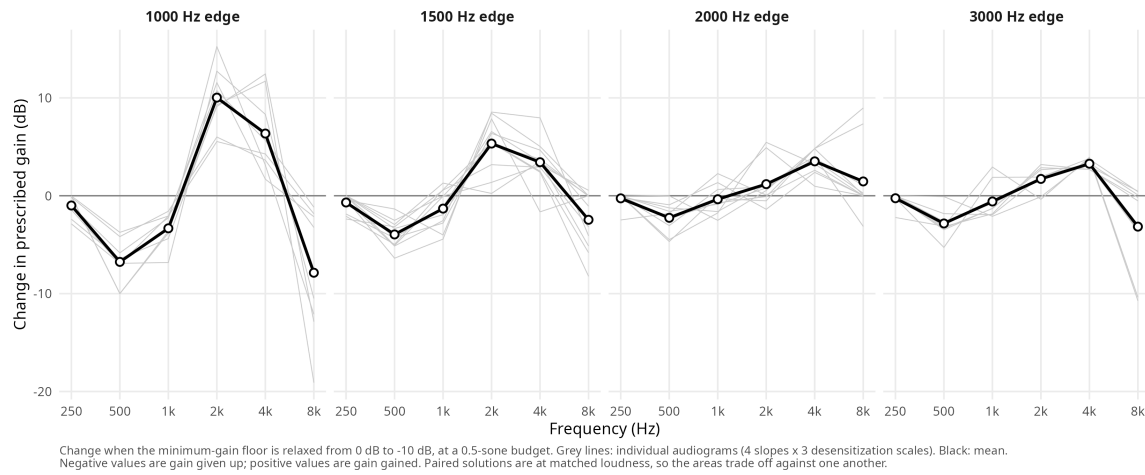


Figure 2: Change in prescribed insertion gain when the minimum-gain floor is relaxed from 0 dB to -10 dB, at a 0.5-sone loudness budget, by audible edge frequency. Grey lines show individual audiograms (four slopes by three desensitization scales); the black line is their mean. Because the paired solutions are at matched loudness, gain given up at one frequency is gain gained at another.

Formula Evaluation: Iso-Loudness Control

To isolate the exact SII cost of preventing low-frequency attenuation, an iso-loudness control experiment was conducted. The Open-NL intelligibility maximizer was constrained to match the *exact* mathematical loudness output generated by NAL-NL2 for each profile at 65 dB SPL. It was then optimized twice: once with a strict 0 dB low-frequency insertion gain floor, and once with a -10 dB floor.

During optimization, Open-NL maximizes a modified SII incorporating a smoothed implementation of the Johnson and Dillon (2011) (Johnson and Dillon 2011) desensitization penalty, aligning its internal objective function closely with the physiological assumptions of NAL-NL2. To ensure rigorous evaluation, both formulas were scored in Table 3 using the smoothed Johnson and Dillon (2011) desensitized SII metric. This aligns the reported index with the intelligibility term of the optimizer's objective, though not with the penalized objective it actually maximizes, so that measured changes reflect the mathematical boundaries rather than a mismatch in the intelligibility metric itself. The primary goal is not to claim algorithmic superiority over the clinical rationale,

but to isolate the exact efficiency cost of the 0 dB insertion floor within a rigidly controlled mathematical space.

Table 3 details the difference in achieved desensitized SII when both the formula and the optimizer are constrained to the same total sones. By separating the *optimizer effect* (the difference between Open-NL at 0 dB and NAL-NL2) from the *floor effect* (the difference between Open-NL at -10 dB and 0 dB), the precise mechanism of loudness crowding-out becomes clear.

Table 3. Iso-Loudness Control: Smoothed Desensitized SII (Constrained to NAL-NL2 Loudness at 65 dB SPL)

Profile	NAL-NL2 Loudness	NAL-NL2 SII	Open-NL (0 dB)	Open-NL (-10 dB)	Optimizer Effect	Floor fect	Ef-
A1 (Mild)	4.14 sones	0.734	0.753	0.786	+0.019	+0.033	
A2 (Re-verse Slope)	3.51 sones	0.769	0.788	0.799	+0.019	+0.011	
A3 (Moderate Sloping)	3.63 sones	0.608	0.605	0.678	-0.003	+0.073	
A4 (Moderate Severe Precipitous)	6.56 sones	0.670	0.699	0.719	+0.029	+0.020	
A5 (Profound Precipitous)	5.50 sones	0.539	0.586	0.595	+0.047	+0.009	
A6 (Mixed)	2.10 sones	0.717	0.719	0.719	+0.003	0.000	
A7 (Conductive)	3.10 sones	0.971	0.976	0.976	+0.005	0.000	

Scoring with the desensitized SII reveals a strikingly different decomposition than raw ANSI SII would suggest, and one that is far more clinically realistic. The floor effect is highly variable, with the moderate sloping profile A3 showing the largest benefit (+0.073) from relaxing the low-frequency floor, followed by the mild profile A1 (+0.033). For the precipitous profiles, the floor effect is modest for A4 (+0.020) and smaller though still measurable for A5 (+0.009). The desensitized SII improvement over NAL-NL2 for these profound losses instead comes substantially from the optimizer effect: +0.029 for A4 and +0.047 for A5. Because the desensitization penalty heavily discounts audibility in the profoundly impaired high frequencies, freeing up additional loudness capacity via low-frequency attenuation yields limited marginal desensitized SII benefit for the steepest high-frequency losses — much of the extra gain is prescribed into frequency regions where the Johnson and Dillon (2011) penalty renders it less valuable.

This result has two important implications. First, it confirms that NAL-NL2's conservative high-frequency gain limits are well-aligned with desensitized intelligibility: the gap between NAL-NL2 and the 0 dB optimizer is modest, while relaxing the floor adds a further benefit that varies substantially across profiles and is largest where high-frequency loss is least severe. Second, the structural loudness crowding-out documented in the audiogram family sweep remains a genuine acoustic boundary, but its *clinical* impact as measured by desensitized SII is substantially attenuated by the very physiological limits that motivated the desensitization correction. The crowding-out constraint is most consequential for profiles where high-frequency loss is moderate enough that the desensitization penalty is small — precisely the profiles (like A3) where the floor effect is largest. Supporting this, the benefit of relaxing the floor shrinks as the desensitization penalty strengthens: for A3 the floor effect is +0.094 with no desensitization, +0.073 smoothed, and +0.045 complete, while for A5 it is +0.006, +0.009, and −0.002 across the three treatments.

For the mixed and conductive profiles (A6, A7) the floor effect is exactly zero and no frequency band receives negative gain, so the floor never engages. For A2, the reverse-slope profile, the band receiving negative gain is 4000 Hz rather than the low frequencies —the only such case—which naturally follows from the reverse-slope configuration. As a

caveat, for A1 the two solutions differ in achieved loudness by 0.016 sones, so that profile alone is not exactly iso-loudness; every other profile matches to within 0.002 sones.

Discussion

The $L_{cap} - L_0$ budget bounds the achievable high-frequency gain for precipitous losses as an independent constraint. For a profile like A4 at 65 dB SPL, unamplified speech inherently generates 5.77 sones of loudness. If evaluated against a strict 9.03 sone normal-hearing ceiling, this unamplified energy consumes roughly 64% of the total capacity before a single decibel of prescriptive gain is applied, though this exact proportion is highly dependent on the chosen L_{cap} . While empirical rationales like NAL-NL2 correctly limit high-frequency gain in these profiles due to physiological desensitization (leaving a small fraction of the loudness budget unused), any clinical attempt to aggressively restore high-frequency audibility beyond these conservative limits (e.g., to maximize raw SII) will immediately collide with this budget constraint. When clinical software enforces a minimum 0 dB insertion gain floor, that residual third of the budget must cover the entire high-frequency prescription, and it is nowhere near sufficient for thresholds of 70 to 100 dB HL.

This constraint forces a harsh physical reality in clinical practice. The penalized-optimal solutions in the audiogram family sweep rely on negative low-frequency insertion gain to free up loudness capacity. However, digital gain reduction cannot bring the ear canal level below the direct sound path in an open fitting: an open earset provides no measurable insertion loss (Kuk, Keenan, and Lau 2009), and venting reduces amplified low-frequency output rather than the unamplified direct sound (Stuart et al. 1999). To achieve true negative low-frequency insertion gain, a clinician must use an unvented earmold, since venting largely removes passive low-frequency attenuation (Kuk, Keenan, and Lau 2009). Even unvented molds attenuate only modestly at the lowest frequencies, and they produce the largest occlusion effects of any configuration (Kuk, Keenan, and Lau 2009; Kiessling et al. 2005), a particular burden for precipitous profiles with near-normal low-frequency hearing.

Conversely, if the clinician opts for an open or vented fitting to avoid occlusion (the standard of care for precipitous losses), the direct sound path locks the low-frequency insertion gain floor at ≥ 0 dB. As demonstrated by the audiogram family

sweep, this instantly traps the fitting at the 0 dB floor, drastically reducing the achievable high-frequency gain before the loudness cap is breached. This theoretical limitation is compounded by practical electroacoustic constraints: open fittings suffer from severe acoustic feedback, which independently limits the maximum stable high-frequency gain a device can deliver.

Crucially, this analysis does not suggest that clinicians should actively pursue these theoretical SII maximums by occluding patients and aggressively attenuating low frequencies. Due to physiological desensitization and cochlear dead regions (Cox et al. 2011; Pepler, Lewis, and Munro 2015), the standard ANSI SII is known to overpredict actual behavioral speech recognition in profound high-frequency losses. Applying massive high-frequency gain often yields diminishing or even negative behavioral returns for these patients. Rather, this index-level analysis serves to map the absolute boundaries of the acoustic parameter space. It demonstrates that even if a clinician *wished* to pursue higher high-frequency gain—perhaps for a patient with exceptionally robust high-frequency neural survival—they are structurally blocked by the loudness budget arithmetic of the open fitting.

Ultimately, maximizing the theoretical Speech Intelligibility Index through raw high-frequency gain in precipitous losses is fundamentally a direct trade-off against residual low-frequency hearing, and achieving it physically requires intolerable occlusion. This structural bottleneck perfectly illustrates why alternative signal processing strategies, such as nonlinear frequency compression or transposition, are so critical (Simpson 2009). By shifting high-frequency speech cues into lower-frequency regions, these algorithms bypass the need for massive high-frequency gain, circumventing both the broadband loudness budget and the limits of high-frequency physiological desensitization entirely.

Limitations

Several methodological constraints should be noted when interpreting these computational results. First, the specific-loudness engine utilizes a monaural model; it does not explicitly capture the complex dynamics of bilateral loudness summation in impaired listeners. Second, the framework applies stationary loudness integration to a static long-term average speech spectrum (LTASS). Real-world speech is highly time-varying, and dynamic compression systems (WDRC) acting on fluctuating speech may yield different

instantaneous loudness profiles (Souza 2002). Furthermore, the analysis is fundamentally a theoretical, index-level optimization; it lacks behavioral validation and does not directly measure patient speech recognition outcomes. Finally, although the analysis evaluated the seven canonical profiles defined by Johnson and Dillon (2011) alongside a systematic sweep of sixteen synthetic audiograms, the simulations were deliberately constrained to synthetic thresholds and a single standard speech spectrum. A broader corpus of real-world audiometric profiles and variable speech inputs would still be required to generalize these constraints across the diverse clinical population.

Similarly, the flat -10 dB minimum insertion gain floor utilized in these simulations likely overstates achievable attenuation in the lowest frequency bands. The reported floor effects should therefore be read as likely optimistic estimates of the benefit of passive attenuation. A frequency-shaped floor based on measured insertion loss would be more realistic for future modeling, though measured attenuation values are highly sensitive to clinical factors such as earmold seal, insertion depth, and slit leak.

The penalized objective introduces a bias in the opposite direction. The anchor penalty charges 0.1 objective points per dB of shift away from the Open-NL heuristic prescription, which the optimizer floors at 0 dB before shifts are applied, so driving two low-frequency bands to the -10 dB floor costs at least 0.02 in SII-equivalent units. For the profiles with the smallest floor effects (A4, +0.020; A5, +0.009) this cost is of the same order as the effect itself, and the reported values may therefore understate what an unpenalized SII maximizer would achieve. The flat floor and the anchor penalty thus bias the reported floor effects in opposite directions, and the reported values should not be read as a one-sided overestimate.

Additionally, the loudness budget framework is limited in its application to the mixed (A6, 30 dB conductive component) and conductive (A7, 50 dB conductive component) profiles. Because unaided speech is inaudible at conversational levels—only becoming audible at 80 dB SPL with 0.17 and 0.28 sones—“budget above unaided” degenerates to an absolute ceiling for these profiles. Evaluated against the 9.03-sone ceiling at 65 dB SPL, both NAL-NL2-aided (2.10 and 3.10 sones; fractions of 0.23 and 0.34) and Open-NL-aided prescriptions (3.46 and 3.84 sones; fractions of 0.38 and 0.42) use a smaller fraction of the budget than for any sensorineural profile (A1–A5 range 0.39 to 0.73). This

is consistent with attenuation without recruitment, and the binding constraint for mixed and conductive losses is therefore the air-bone-gap anchoring rather than the loudness cap.

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Data Availability

The code used to execute the computational simulations, reproduce the dataset, and generate the figure for this study is fully open-source and available on GitHub (<https://github.com/r-gregmisc/SII>).

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